

START

READ MealQuantity, DrinkQuantity, Payment

$\text{MealSubtotal} = \text{MealQuantity} * 85.00$

$\text{DrinkSubtotal} = \text{DrinkQuantity} * 25.00$

$\text{MealDiscount} = \text{MealSubtotal} * 0.10$

$\text{DiscountedMealTotal} = \text{MealSubtotal} - \text{MealDiscount}$

$\text{PackagingFee} = \text{MealQuantity} * 5.00$

$\text{FinalBill} = \text{DiscountedMealTotal} + \text{DrinkSubtotal} + \text{PackagingFee}$

$\text{Change} = \text{Payment} - \text{FinalBill}$

DISPLAY (MealSubtotal)

DISPLAY (DrinkSubtotal)

DISPLAY (MealDiscount)

DISPLAY (PackagingFee)

DISPLAY (FinalBill)

DISPLAY(Change)

END

Test Case 1 (Meal = 3, Drink = 4, Payment = ₱500.00)

- MealSubtotal = $3 * 85.00 = 255.00$
- DrinkSubtotal = $4 * 25.00 = 100.00$
- MealDiscount = $255.00 * 0.10 = 25.50$
- PackagingFee = $3 * 5.00 = 15.00$
- FinalBill = $(255.00 - 25.50) + 100.00 + 15.00 = 344.50$
- Change = $500.00 - 344.50 = 155.50$

Outputs:

Meal Subtotal: ₱255.00

Drink Subtotal: ₱100.00

Discount: ₱25.50

Packaging Fee: ₱15.00

Final Bill: ₱344.50

Change: ₱155.50

Test Case 2 (Meal = 5, Drink = 3, Payment = ₱600.00)

- MealSubtotal = $5 * 85.00 = 425.00$
- DrinkSubtotal = $3 * 25.00 = 75.00$
- MealDiscount = $425.00 * 0.10 = 42.50$
- PackagingFee = $5 * 5.00 = 25.00$
- FinalBill = $(425.00 - 42.50) + 75.00 + 25.00 = 482.50$
- Change = $600.00 - 482.50 = 117.50$

Outputs:

Meal Subtotal: ₱425.00

Drink Subtotal: ₱75.00

Discount: ₱42.50

Packaging Fee: ₱25.00

Final Bill: ₱482.50

Change: ₱117.50

Question: Why would applying the 10% discount to the entire bill produce an incorrect result?

Answer: Applying the 10% discount to the entire bill would incorrectly reduce the price of the drinks and packaging fees. The canteen's rule only gives 10% discounts only to the rice meals while drinks and packaging fees must be charged with full price.